```
# systemctl enable puppet.service.
```

4. AIDE

System stability is put at risk when configuration files are deleted or modified without authorisation or careful supervision. How can a change to an important file or directory be detected? This problem can be solved by using intrusion detection software to monitor files for changes. Advance Intrusion Detection Environment or AIDE can be configured to monitor files for a variety of changes including permissions or ownership changes, timestamp changes, or content changes.

To get started with AIDE, install the RPM package that provides the AIDE software. This package has useful documentation on tuning the software to monitor the specific changes of interest.

```
# yum -y install aide.
```

Once the software is installed, it needs to be configured. The */etc/aide.conf* file is the primary configuration file for AIDE. It has three types of configuration directives: configuration lines, selection lines and macro lines.

Configuration lines take the form *param=value*. When *param* is not a built-in AIDE setting, it is a group definition that lists which changes to look for. For example, the following group definition can be found in */etc/aide.conf*, which is installed by default:

```
PERMS = p+i+u+g+acl+selinux
```

The aforesaid line defines a group called PERMS that looks for changes in file permissions *(p)*, inode *(i)*, user ownership *(u)*, group ownership *(g)*, ACLs *(acl)*, or SELinux context *(selinux)*. We can write our own parameters in the file which can be used to monitor our systems.

Selection lines define which checks are performed on matched directories.

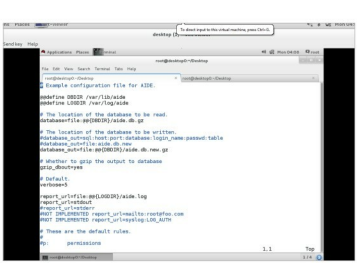

Figure 3: The configuration file of AIDE */etc/aide.conf*

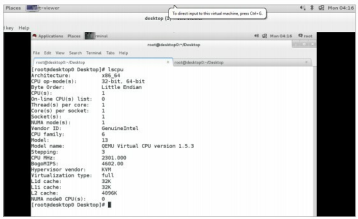

Figure 4: CPUs running in a machine can be identified by the *lscpu* command

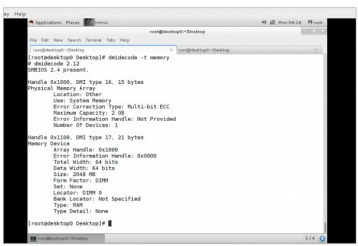

Figure 5: The *dmidecode* command in use

The third type of directive is macro lines — these define variables and their definition has the following syntax.

```
@@define VAR value
```

Execute the *aide --init* command to initialise the AIDE database. After the database is created, a file system check can be performed using the *aide --check* command. This command scans the file system and compares the current state of files with the information in the earlier AIDE database. Any differences that are found will be displayed, showing both the original file's state and the new condition.

5. Mcelog

An important step in troubleshooting potential hardware issues is knowing exactly which hardware is present in the system. The CPU(s) in a running system can be identified from the *lscpu* command, as shown in Figure 4.

The *dmidecode* tool can be used to retrieve information about physical memory banks, including the type, speed and location of the bank as shown in Figure 5.

Modern systems can typically keep a watch on various hardware failures, alerting an administrator when a hardware fault occurs. While some of these solutions are vendor-specific and require a remote management card, others can be read